CO3-AT3

4] A fixed-point DSP implementation gives an SQNR of 32 dB for a 10-bit quantizer. Determine whether the result is correct and debug the computation.

PROGRAM;

```
clc;
clear;
close all;

%% Parameters
N = 10;        % 10-bit quantizer
A = 1;         % Signal amplitude
fs = 10000;
f = 1000;

t = 0:1/fs:0.01;

%% Generate sine wave
x = A*sin(2*pi*f*t);

%% Quantizer step size
Delta = (2*A)/(2^N);

%% Quantization
xq = Delta*round(x/Delta);

% Limit to input range
xq(xq > A) = A;
```

```
xq(xq < -A) = -A;

%% Quantization error
e = x - xq;

%% Signal and noise power
Ps = mean(x.^2);
Pq_sim = mean(e.^2);

%% Theoretical noise power
Pq_theory = Delta^2/12;

%% SQNR
SQNR_sim = 10*log10(Ps/Pq_sim);
SQNR_theory = 6.02*N + 1.76;

%% Display results
fprintf('Number of bits           = %d\n', N);
fprintf('Quantization step size    = %.9f V\n', Delta);
fprintf('Signal power             = %.6f V^2\n', Ps);
fprintf('Simulated noise power     = %.9e V^2\n', Pq_sim);
fprintf('Theoretical noise power   = %.9e V^2\n', Pq_theory);
fprintf('Simulated SQNR           = %.2f dB\n', SQNR_sim);
fprintf('Theoretical SQNR         = %.2f dB\n', SQNR_theory);

%% Compare with reported value
reported_SQNR = 32;
```

```
fprintf('\nReported SQNR          = %.2f dB\n', reported_SQNR);
fprintf('Expected SQNR          = %.2f dB\n', SQNR_theory);
fprintf('Difference             = %.2f dB\n', ...
    SQNR_theory - reported_SQNR);

if abs(reported_SQNR - SQNR_theory) < 1
    fprintf('Result is CORRECT.\n');
else
    fprintf('Result is INCORRECT for an ideal 10-bit quantizer.\n');
end
```

OUTPUT;

```
Number of bits                 = 10
Quantization step size         = 0.001953125 V
Signal power                   = 0.495050 V^2
Simulated noise power          = 9.667737931e-09 V^2
Theoretical noise power        = 3.178914388e-07 V^2
Simulated SQNR                 = 77.09 dB
Theoretical SQNR               = 61.96 dB

Reported SQNR                  = 32.00 dB
Expected SQNR                  = 61.96 dB
Difference                     = 29.96 dB
Result is INCORRECT for an ideal 10-bit quantizer.
```